Identification and Control of a Propeller-Actuated Damped Pendulum (PADP)

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Abstract—This paper presents the experimental identification and closed-loop control of a propeller-actuated damped pendulum (PADP). Three step-response datasets were collected and processed using both manual parameter approximation and MATLAB's System Identification toolbox, yielding six candidate transfer functions. Statistical analysis identified one experiment as disturbance-affected and excluded it; the final plant model was obtained by averaging the four remaining models. Three continuous-time controllers were designed — a PID tuned via IMC in SISOtool, an integral state feedback regulator via LQR (ISF-LQR), and an I-PD by pole placement — and physically validated against a settling time of 5 s and maximum overshoot of 3%. The ISF-LQR and I-PD controllers satisfied both specifications, while the PID exhibited the poorest transient response among the three.

I. Introduction

The propeller-actuated damped pendulum (PADP) is an underactuated mechanical system whose angular position is governed by the aerodynamic torque of a brushless motor and propeller assembly. Around a fixed operating point, the nonlinear dynamics admit a linear second-order underdamped approximation [1], making it a representative platform for comparing control strategies under the design requirements of settling time $t_s \leq 5\,\mathrm{s}$ and maximum overshoot $M_p \leq 3\,\%$.

Plant identification was carried out using both parametric methods — extracting ω_n and ζ from measured transient features — and data-driven fitting via MATLAB's ident toolbox [2]. Statistical screening across repeated trials was applied to handle noise and disturbances before model averaging.

Three control architectures were implemented and compared. A continuous PID tuned via Internal Model Control (IMC) [3], a Linear Quadratic Regulator with augmented integral state (ISF-LQR) [4], and an I-PD controller with pole placement [5], where the proportional and derivative terms act on the plant output to eliminate derivative kick and reduce overshoot.

This paper reports the complete identification-to-validation cycle for the PADP. Section II covers parameter estimation and plant derivation. Section III details the controller designs. Sections IV and V present and interpret the experimental results against the design specifications.

II. System Identification

A. Experimental Setup and Data Acquisition

The PADP platform consists of an extruded-aluminium support rigidly fixed to a wooden base. A hollow aluminium arm pivots on bearings at the top of the support and carries a Hobbymate 2204 brushless motor governed by a SimonK 20 A ESC fitted with a GEMFAN 5045BNR three-blade propeller. Angular position is measured by a USDigital E5 incremental encoder (3600 CPR) coupled to the pivot shaft and sampled every 20 ms with a gain $k = 2\pi/3600$ rad/count. PWM actuation uses OneShot125 mode at a 500 µs period. Three independent experiments were conducted by applying a step input of 1.7 units from rest, establishing a steady-state operating point near $\theta_{eq} \approx 0.6$ rad (≈ 34.4), and recording the angular response for 40 s.

B. Parameter Approximation

The logarithmic decrement method [1] was applied to each trial, yielding:

$$\delta = \frac{\ln(y_0/y_i)}{i}, \quad \zeta = \frac{\delta}{\sqrt{(2\pi)^2 + \delta^2}}, \quad \omega_n = \frac{2\pi/T}{\sqrt{1-\zeta^2}} \tag{1}$$

yielding three second-order transfer functions of the prototype:

$$G_i(s) = K_i \frac{\omega_{n,i}^2}{s^2 + 2\zeta_i \omega_{n,i} s + \omega_{n,i}^2}, \quad i = 1, 2, 3 \tag{2}$$

TABLE I: Parameters estimated by the logarithmic decrement method

Trial	K	ζ	ω_n (rad/s)
1	0.338	0.032	3.9292
2	0.340	0.035	3.9294
3	0.423	0.033	3.9291

C. MATLAB System Identification

MATLAB's ident toolbox fitted the same prototype to each dataset, producing three additional models (Table II).

TABLE II: Parameters estimated by system identification

Trial	K	ζ	ω_n (rad/s)
1	0.346	0.0292	3.9225
2	0.346	0.0295	3.9301
3	0.426	0.0296	3.9091

D. Statistical Analysis and Model Selection

The six models were pooled and their parameters compared statistically. Models from trial 3 deviated by more than $2s$ from the group mean, consistent with an external disturbance, and were excluded. The final plant was obtained by averaging the remaining four models:

$$G(s) = \frac{5.287}{s^2 + 0.2481s + 15.43} \tag{3}$$

III. Controller Design

Design specifications for all regulators: settling time $t_s \leq 5$ s (2% criterion) and maximum overshoot $M_p \leq 3\,\%$ to a 0.6 rad step reference, with zero steady-state error and disturbance rejection.

A. Continuous PID Controller (SISOtool)

A continuous-time PID controller was designed using the Internal Model Control (IMC) tuning methodology. The IMC parameter was deliberately tightened to achieve $\%OS \approx 0\%$ in simulation, since the plant already introduces an inherent overshoot that could accumulate and violate the 3% bound in real operating conditions. The controller incorporates a filtered derivative action to attenuate high-frequency noise, as given by (5).

$$PID(s) = K_p + \frac{K_i}{s} + K_d\, s \cdot \frac{N}{s+N} \tag{4}$$

«««< HEAD where K_p is the proportional gain, K_i the integral gain, K_d the derivative gain and N the derivative filter coefficient.

Coefficient matching of the IMC-equivalent open-loop transfer function yielded: ======= where K_p is the proportional gain, K_i the integral gain, K_d the derivative gain and N the derivative filter coefficient.The tuning procedure yielded the following parameter values: »»»> Daniel

TABLE III: IMC-tuned PID controller parameters.

K_p	K_i	K_d	N
−0.634	2.000	0.343	3

The negative proportional gain is consistent with the sign convention of the identified plant, which exhibits an inherently inverting static gain.

The PID controller with their values can be written as:

$$C(s) \;=\; -0.634 \;+\; \frac{2}{s} \;+\; 0.343\,\frac{3s}{s+3} \tag{5}$$

B. Integral State Feedback Controller (ISF) via LQR

An Integral State Feedback (ISF) controller was designed using the Linear Quadratic Regulator (LQR) criterion. The method augments the plant with an integral state to ensure zero steady-state error and improve disturbance rejection.

The identified plant in controllable canonical form is:

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} -0.2481 & -15.4300 \\ 1 & 0 \end{bmatrix}, & \mathbf{B} &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \\ \mathbf{C} &= \begin{bmatrix} 0 & 5.2870 \end{bmatrix}, & \mathbf{D} &= 0 \end{aligned} \tag{6}$$

The system is fully controllable (rank = 2). To include integral action, the augmented model is defined as:

$$\dot{x}_I = r - \mathbf{C}\mathbf{x} \tag{7}$$

$$\mathbf{A}_a = \begin{bmatrix} -0.2481 & -15.4300 & 0 \\ 1 & 0 & 0 \\ 0 & -5.2870 & 0 \end{bmatrix}, \quad \mathbf{B}_a = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \tag{8}$$

The augmented system is also controllable (rank = 3). Based on the specifications ($t_s \leq 5$ s, $M_p \leq 3\,\%$), the desired poles were selected as:

$$p_{1,2} = -2.8 \pm 3.1266j, \quad p_3 = -14.0 \tag{9}$$

The LQR weighting matrices were chosen using Bryson's rule:

$$\mathbf{Q} = \operatorname{diag}\left(\frac{1}{9^2}, \frac{1}{7^2}, \frac{1}{0.45^2}\right), \quad R = \frac{1}{0.6^2} \tag{10}$$

The resulting gain vector $\hat{\mathbf{K}} = [k_1 \;\; k_2 \;\; K_i]$ is:

$$\hat{\mathbf{K}} = \begin{bmatrix} 0.7214 & 0.4370 & -1.3333 \end{bmatrix} \tag{11}$$

which yields the closed-loop poles:

$$p_{1,2} = -0.2593 \pm 3.9454j, \quad p_3 = -0.4509 \tag{12}$$

C. I-PD Controller via Pole Placement

The I-PD structure places the integrator on the error signal while the proportional and derivative terms act on the plant output, preventing derivative kick on step changes of the reference:

$$C_{\text{I-PD}}(s) = \frac{K_i}{s} - (K_p + K_d\, s) \tag{13}$$

The target specifications ($t_s \leq 5$ s, $M_p \leq 3\,\%$) were mapped to the damping ratio ζ_d and natural frequency $\omega_{n,d}$ of the desired dominant poles. A third real pole was placed sufficiently far to the left to remain non-dominant. Coefficient matching of the closed-loop characteristic polynomial yielded K_p, K_d, and K_i, subsequently decomposed into P, I, D, N parameters for the PASCAL PID block. The controller was designed to specifications more strict than the nominal requirements ($t_s \leq 2.9$ s, $M_p \leq 1\,\%$) to provide a performance margin that compensates for the degradation expected during physical implementation.

$$p_{1,2} = -1.3793 \pm j0.4705 \quad p_3 = -6.8966$$

$$K_p = 1.0817, \quad K_i = 2.7704, \quad K_d = 1.7793, \quad N = 20$$

The third real pole p_3 was placed at five times the magnitude of the real part of the dominant poles, following the standard non-dominance criterion [1]:

$$|p_3| = 5\,|\mathrm{Re}(p_{1,2})| = 5 \times 1.3793 = 6.897$$

This placement ensures that the transient contribution of p_3 decays approximately five times faster than the dominant pair

D. Discrete PID Controller (SISOtool)

A discrete-time PID controller was designed through manual pole-zero placement in the z-domain, without resorting to an automated tuning method such as IMC. The compensator structure was chosen to consist of two real poles and two complex conjugate zeros, which corresponds to the discrete equivalent of a PID action with filtered derivative. The design procedure consisted of iteratively adjusting the controller gain, the natural frequency associated with the complex zero pair, and the pole locations until the closed-loop step response satisfied the specifications.

The resulting discrete controller transfer function is:

$$C(z) = \frac{3.0189\,(z^2 - 1.95z + 0.9881)}{(z - 0.09)(z - 1)} \tag{14}$$

IV. Results and Analysis

This section presents the results obtained in both simulation and physical implementation for each controller, evaluating compliance with the performance specifications.

A. Continuous PID Controller

The figure 1 shows the simulated closed-loop step response obtained by connecting the designed controller around the identified plant model.The output settles within the ±2% band with the response oscillating persistently around the reference once inside it. The sustained oscillation indicates that the simulation is operating at the boundary of the stability margin introduced by the IMC tuning.

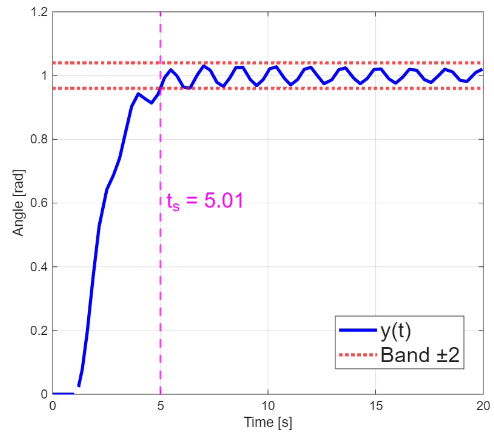

Fig. 1: Simulated step response of the PID controller.

The experimental step response recorded on the physical platform. The out put enters and remains inside the ±2% band with a small peak overshoot before settling cleanly, as summarised in Table IV. A disturbance injected at $t = 20\,\mathrm{s}$ causes a transient excursion, after which the controller successfully rejects it and returns the output to the reference, demonstrating integral action and robustness to step disturbances.

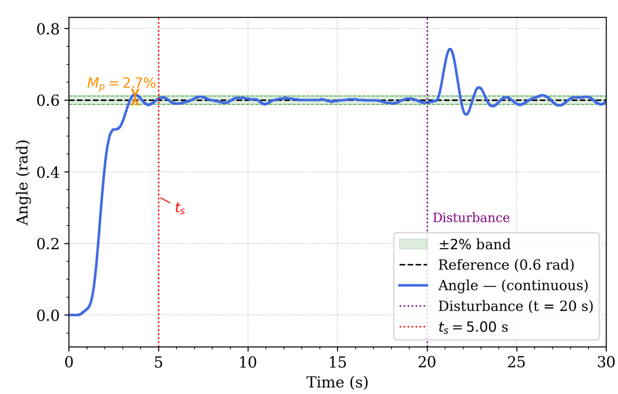

Fig. 2: Experimental step response of the PID controller.

The primary difference between the two responses lies in the steady-state behaviour after settling. In simulation, the output exhibits a sustained low-amplitude oscillation that keeps it cycling inside the ±2% band, whereas experimentally the response damps out and converges to the reference without visible oscillation. Furthermore, the experimental response presents a visible peak overshoot before entering the band, a feature absent in the simulated response.

To improve the experimental behaviour, the derivative filter coefficient was adjusted from $N = 3$ to $N = 5$ while keeping K_p, K_i, and K_d unchanged. Increasing N raises the filter cutoff frequency, allowing the derivative action to respond over a wider frequency range. This produces a stronger damping effect on the real platform, which explains both the suppression of the post-settling oscillation and the cleaner convergence observed experimentally. On the real plant, the increased derivative activity provided by $N = 5$ attenuates those higher-frequency components, resulting in a smoother and more stable steady-state response.

TABLE IV: PID performance metrics: simulation vs. experiment

Condition	t_s (s)	M_p (%)	e_{ss}	Meets specs?
Simulation	5.01	3.4	0	No
Experimental	5	2.7	0	Yes

B. Integral State Feedback Controller via LQR

The closed-loop step responses of the ISF-LQR controller for $r = 0.6$ rad are shown in Fig. 1 (experiment) and Fig. 2 (simulation). The corresponding performance metrics are summarized in Table V.

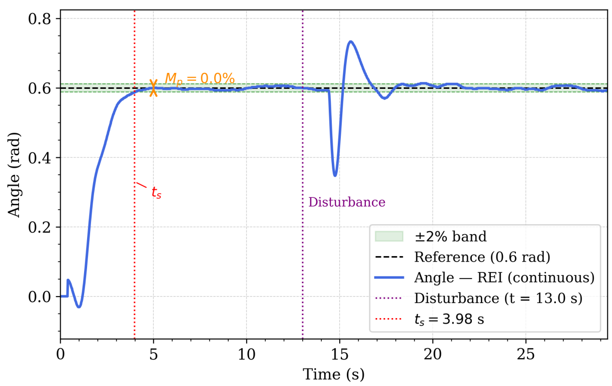

Fig. 3: Experimental step response of the ISF-LQR controller.

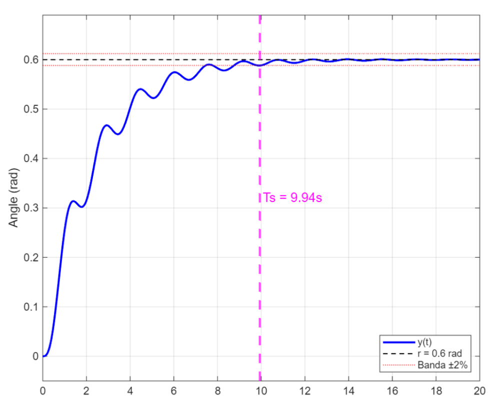

Fig. 4: Simulated step response of the ISF-LQR controller.

TABLE V: ISF-LQR performance metrics: simulation vs. experiment

Condition	t_s (s)	M_p (%)	e_{ss}	Meets specs?
Simulation	9.94	0.12	0	No
Experimental	3.98	0	0	Yes

Both conditions satisfy $e_{ss} = 0$ and $M_p \leq 3\%$; the settling time discrepancy is addressed in Section V.

C. I-PD Controller via Pole Placement

The closed-loop step responses for the I-PD controller are shown in Figs. 5 (experimental) and 6 (simulation). Performance metrics are summarized in Table VI.

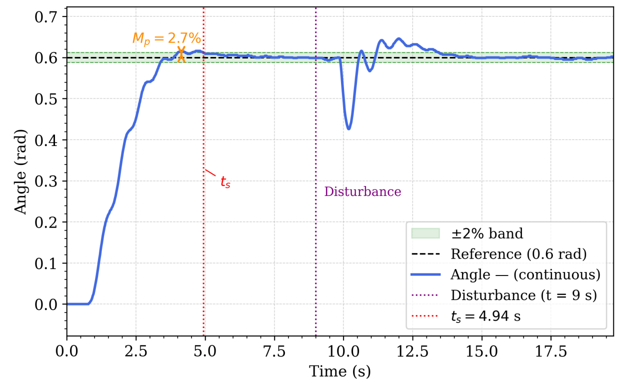

Fig. 5: Experimental step response of the I-PD controller.

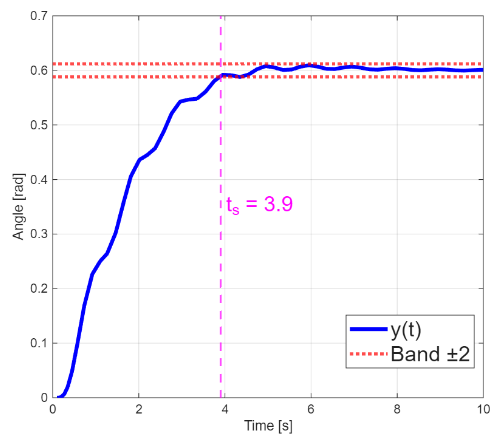

Fig. 6: Simulated step response of the I-PD controller.

The simulated response exhibits near-critically-damped behaviour ($M_p = 1\%$, $t_s = 3.9$ s), consistent with the chosen pole locations ($\zeta_d \approx 0.946$). The actuator saturation included in simulation produces a mild oscillatory entry visible in both curves, caused by the motor saturating to zero when the pendulum exceeds the reference, a physically inevitable consequence of the unidirectional propeller thrust. Despite this, both responses remain within specifications.

TABLE VI: I-PD performance metrics: simulation vs. experiment

Condition	t_s (s)	M_p (%)	e_{ss}	Meets specs?
Simulation	3.9	1.0	0	Yes
Experimental	4.94	2.7	0	Yes

D. Discrete PID Controller

The discrete PID controller (15), designed via manual pole-zero placement, failed to meet the imposed specifications on the physical plant. Unlike the continuous IMC design, where the tuning parameter provides a structured and systematic path to satisfy the closed-loop requirements, the manual approach is inherently sensitive to unmodelled dynamics. In discrete time, this sensitivity is further amplified by quantisation effects introduced by the sampling process, discretisation approximation errors in the derivative action, and the reduced phase margin that results from the computational delay added by the digital implementation. These factors, absent in the continuous design, cause the discrete controller to behave differently on the real platform than predicted by the linear model. Due to the unsatisfactory experimental outcome, no simulation or experimental plots are included for this controller.

E. Comparative Summary

V. Discussion

This section interprets the results presented above, analyzing the causes of the observed differences between theoretical models and the physical behavior of the system, as well as the advantages and disadvantages of each implemented control strategy.

A. ISF-LQR: Simulation–Experiment Discrepancy

Figs. 4 and 5 show a clear contrast between simulated and experimental responses. The simulation exhibits a slowly damped transient with $t_s = 9.94$ s, consistent with the real pole $p_3 = -0.4509$ near the imaginary axis, which dominates the settling behavior despite the faster complex poles. In contrast, the experimental response converges monotonically to the 0.6 rad reference in $t_s = 3.98$ s with zero overshoot, satisfying both specifications.

A small negative deflection at $t \approx 1$ s in the experimental response (about -0.07 rad) is attributed to actuator dead-zone and propeller inertia, which delay the onset of effective control action. This effect is not captured by the linear model.

The faster experimental convergence is consistent with additional nonlinear damping (e.g., friction and aerodynamic drag), which increases the effective damping and suppresses oscillations, leading to a shorter settling time than predicted.

Finally, the experimental response shows effective disturbance rejection; a perturbation at $t \approx 13$ s produces a deviation of approximately 0.25 rad, after which the output returns to the reference, remaining within the margin of less than 2% steady-state error. This confirms the role of comprehensive action in rejecting disturbances despite the model's inaccuracies.

VI. Conclusions

The ISF-LQR controller achieved zero steady-state error and effective disturbance rejection. While the simulation resulted in $t_s = 9.94$ s and $M_p = 0.12\,\%$, failing the settling time requirement, the experimental response reached $t_s = 3.98$ s with $M_p = 0\,\%$, satisfying all specifications.

References

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